

Problem Set #21

September 2025

Problem 1. You're creating a new coin system for your country. You must use only four coin values and you must be able to create the values 1 through 10 using one coin at a minimum and two coins maximum. What 4 coins do you choose, and can you think of a second set of 4 coins that achieves the same goal?

Problem 2. The game starts with 6 empty glasses in a row numbered 1 to 6. You roll a standard die. If the number for the glass is empty, then the glass is filled up. If the number for the glass is full, then you drink that glass so it becomes empty.

If all 6 glasses are full, the game ends and you drink them all. How much do you drink, on average?

Problem 3. Let $a_1, a_2, \dots, a_n$ be a sequence of integers. Show that there exist integers j and k with $1 \leq j \leq k \leq n$ such that the sum $\sum_{i=j}^k a_i$ is a multiple of n .

Problem 4. Show that the integral of $\int_0^1 \frac{x^n}{(x^2+1)} dx$ is irrational for all nonnegative integers n . You may use the fact that π and $\ln(2)$ are irrational.

Problem 5. There are 100 people on an island. 33 of them have green eyes, but nobody can see if their own eyes are green (no mirrors etc). each day, they stand in a circle and look at each other. no communication is allowed at any time. if someone knows that they have green eyes, they must leave the island. after how many days does the last green-eyed person leave?